Biochem-III
15/1/0

QP Code : 12851

(2½ Hours) [Total Marks : 75

N.B. : (1) **All** the questions are **compulsory**. Choice is **internal.**
(2) **Figures** to the **right** indicate **full** marks.
(3) **All** questions carry **equal** marks.
(4) Draw **flowcharts/diagrams** wherever **necessary**.

1. (A) Choose the MOST appropriate option : (any **three**)
(i) The elongation step of replication uses __________ as a source of energy. (dNTP / rNTP / only ATP)
(ii) ____________ is required for the process of initiation of replication. (SSB / Primosome / Replisome
(iii) DNA synthesis takes place in ___________ direction (3'-5'/ 5'-3'/ 3'-5'or 5'-3')
(iv) During mismatch repair, the sequence ___ is used for identification of the mismatched base pairs. (CTAG / GAUC / GATC)
(v) UV radiations lead largely to the formation of _________________ . (thymine monomers / thymine dimers / thymine trimers)
(vi) Gyrase helps in _________________. (Unwinding / removing supercoils / joining of Okazaki fragments)

(B) Answer the following any **one** :— 2
(i) Describe the rolling circle mode of DNA replication.
(ii) Give the role of alkyl transferase in DNA repair.

(C) Write short note on any **one** :— 4
(i) Meselson Stahl experiment (ii) Recombination repair

(D) Attempt any **one** :— 6
(i) Discuss semi-discontinuous DNA replication.
(ii) Enlist any two types of DNA damages that usually occur. Elaborate on the Excission repair.

2. (A) Choose the MOST appropriate option : (any **three**) 3
(i) Translation occurs in the ____________. (cytoplasm / nucleus / mitochondria)
(ii) In eukaryotes, a mature RNA transcript that is translated contains __________. (both introns and exons / only introns / only exons).
(iii) ________ box generally acts as promoter at -10 bp during transcription. (CAAT / TATA / either CAAT or TATA)
(iv) The anticodon to the mRNA codon 5'AUG is _______________. (5'UAC / 5' CAU / 5' TAC)
(v) Shine Dalgarno sequence is present on ________________. (16S RNA / mRNA / tRNA)
(vi) ______________________ is a post translational modification. (5' capping / 3' tailing / acetylation)

[TURN OVER

(B) Define / Explain the terms any **one** :— 2

(i) ORF (ii) Anti-sense strand

(C) Write short note on any **one** of the following :— 4

(i) Wobble hypothesis and Genetic Code

(ii) RNA polymerase

(D) Answer in detail any **one** :— 6

(i) Elaborate on the initiation and termination steps in translation.

(ii) Discuss the various modifications that take place post-transcription.

3. (A) Choose the MOST appropriate option : (any **three**) 3

(i) A bacterium protects its DNA from cleavage by its own restriction nucleases as ____________________.
(DNA ligase quickly re-seals target sequences when they are cut / Proteins attach to the target sequences, preventing restriction nucleases from binding / bacterium chemically modifies its own DNA sequences, thereby preventing recognition by the nucleases)

(ii) In Klenow fragment ____________ activity is absent.
(3'-5' exonuclease / 5'-3' exonuclease / 5'-3' polymerase)

(iii) Type II restriction endonuclease has ____________ recognition site.
(asymmetric / palindromic / nonspecific)

(iv) ____________ DNA is generally used as probes.
(Centromere / Telomere / Satellite)

(v) YAC is ____________________ vector.
(only a cloning / only a shuttle / both a cloning and shuttle)

(vi) Eco RI produces ____________________ ends.
(blunt / 3' protruding / 5' protruding)

(B) Define and explain any **one** :— 2

(i) Ti Plasmids (ii) Terminal transferase

(C) Write brief notes on any **one** :— 4

(i) Molecular scissors (ii) Shuttle vectors

(D) Attempt any **one** :— 6

(i) Elaborate on cloning vectors.

(ii) Discuss how genetic engineering has helped revolutionize food and insulin production

4. (A) Choose the MOST appropriate option : (any **three**) 3

(i) Microinjection can be used in ____________________.
(walled cells / protoplast / both walled cells and protoplast)

(ii) The enzyme ____________ can work at high temperatures.
(DNA PolI / Kornberg / Taq polymerase)

[TURN OVER

(iii) Southern blotting is used for separation of ____________.
(DNA / RNA / protein)

(iv) $CaCl_2$ helps in ____________________.
(transformation / PCR / microinjection)

(v) In Lac selection, cells transformed with vectors containing recombinant DNA will produce ___________ colonies. (white / blue / clear)

(vi) PCR was developed by ____________________.
(Watson & Crick / Kary Mullis / University of California)

(B) Define and explain the terms any **one** :— 2
(i) RFLP (ii) Gene transfer

(C) Write short note on any **one** :— 4
(i) Electroporation (ii) Antibiotic selection

(D) Elaborate on any **one** :— 6
(i) PCR (ii) c-DNA library

5. (A) Answer any **one** :— 3
(i) Give the similarities and differences between the various types of DNA Polymerases.
(ii) Discuss the role of ligase and helicase in DNA replication.

(B) Write brief notes on any **one** :— 3
(i) Inhibitors of translation
(ii) Charging of tRNA

(C) Answer in brief any **one** :— 3
(i) What is role of reverse transcriptase in recombinant DNA technology ?
(ii) Write a note on radioactive probes.

(D) Answer in brief on any **one** :— 3
(i) Write a brief note on transfection.
(ii) Schematically only represent the steps in involved in southern blotting technique.

(E) State **true** or **false** (any **three**) :— 3
(i) DNA replication is discontinuous on both strands.
(ii) Okazaki fragments are formed in the leading strand.
(iii) Hind III produces blunt ends.
(iv) Genomic DNA is the starting material for making a cDNA library.
(v) Rho factor is involved in termination of translation.
(vi) Plasmids exhibit the lytic and the lysogenic phases of life cycle.

———v———